

Demotivation in English learning of Chinese medical students: Insights from situated expectancy-value theory

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ABSTRACT

Despite extensive research on motivation in English as a Foreign Language (EFL) settings, demotivation within medical education remains underexplored. This mixed-method study employs the Situated Expectancy-Value Theory (SEVT) to investigate the demotivation of English learning experienced by 426 Chinese medical students in their English language learning. Utilising data collected from the adapted English learning demotivation questionnaire, quantitative analysis was conducted through exploratory factor analysis, confirmatory factor analysis, and correlation analysis. Qualitative insights from open-ended responses revealed that the interplay of internal (e.g., lack of self-regulation) and external (e.g., heavy medical course load) factors plays an important role in medical students' demotivation in English. Significant demotivators identified included weak language foundation, heavy workload of medical courses, and lack of self-regulation. A moderate correlation was found between English proficiency and demotivation, emphasising the importance of considering a range of factors, rather than attributing demotivation solely to low English proficiency. The findings extend the understanding of demotivation and reconceptualise the concept through low expectancy of success, reduced value, and high perceived effort cost. Critical insights emphasise the importance of fostering motivating English learning environment in medical education by considering both individual and contextual factors. Implications for educational interventions are discussed.

1. Introduction

In medical education systems within non-English-speaking countries, undergraduate and postgraduate students are often mandated to undertake English as a Foreign Language (EFL) courses to enhance English proficiency for professional career development (Liu et al., 2020). English language education at the tertiary level aims to foster general communicative competence (English for General Purpose) and language skills for academic engagement (English for Academic Purposes, EAP). In medical schools, EFL courses are further designed and delivered to prepare students for professional academic communication, such as reading and summarising academic literature, publishing journal articles, and attending international conferences (Cheng, 2016). Professional language requirements and critical communication in medical context have made English vital for the professional success of Chinese medical students. Although motivation of EFL learners has been a focal point in language acquisition research for decades, medical students may also experience demotivation, which calls for further attention.

Exploring language learners' demotivation enables instructors to improve their teaching strategies, enrich students' learning experiences, and remotivate learners who have lost interest (Liu, 2022). In motivation studies, Situated Expectancy-Value Theory (SEVT) integrates social cognitive and sociocultural perspectives, formulating a comprehensive framework to understand student motivation by considering expectancy for success, subjective task value, and cost within situated contexts (Eccles & Wigfield, 2020). By applying SEVT, this research seeks to provide an in-depth understanding of demotivation and offer insights for developing effective remotivating strategies.

Research on medical students predominantly focus on psychology and education, particularly on factors influencing their general learning attitudes (Alsoufi et al., 2020), academic performance, self-efficacy, emotions, and relationships between these factors (Hayat et al., 2020; Wu et al., 2020). Recent studies have also explored the psychological impacts and learning stress caused by the COVID-19 pandemic (Abdulghani et al., 2020; Alsoufi et al., 2020). English learning for medical students in non-English speaking countries is vital for their

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professional development and future career preparation. As a highly professional group, medical students in China need to publish research or engage in professional communication in English at international conferences for either postgraduate academic training or doctor promotion (Deng et al., 2022). Medical students have received relatively little attention in the field of demotivation compared to high school students (Gao et al., 2022; Sakai & Kikuchi, 2009; Song & Kim, 2017) or undergraduate students in other majors (Falout et al., 2009; Hassaskhah et al., 2015; Wang & Littlewood, 2021), particularly in the context of EFL learning in non-English-speaking countries (Rabadi & Rabadi, 2020). Beyond limited research focus, English learning among medical students involves complex interdisciplinary interactions of core medical sciences such as anatomy, physiology, and pharmacology, alongside medical ethics and social sciences, all interwoven with language. The emotional toll of medical training can intensify stress and demotivation. Combining these unique disciplines and social-emotional needs, medical students deserve greater attention in their internal drive in English learning. To address these gaps, this study investigates the demotivation of Chinese medical undergraduates in EFL learning through the lens of the Situated Expectancy-Value Theory (SEVT) to explore the interplay of internal and external demotivators (Eccles & Wigfield, 2020).

This study innovatively applies Situated Expectancy-Value Theory (SEVT) to an under-explored cohort: Chinese medical students learning English as a foreign language. It addresses a gap in the literature by examining English learning demotivation in the medical education context. The study provides novel insights into how academic workload and self-regulation challenges demotivate medical students in English learning and how the interplay of internal and external factors, from the SEVT perspective, can facilitate their remotivation.

2. Literature review

2.1. Demotivation and remotivation in language learning

Demotivation is derived from the term amotivation, and both are described as the reduction or diminishment of motivation. Amotivation emphasises the role of perceived meaninglessness as a key factor leading to a lack of motivation among language learners (Deci & Ryan, 2013). Demotivation highlights specific external forces that reduce or eliminate the drive for language learning behaviours (Christophel & Gorham, 1995; Dörnyei & Ushioda, 2021). Demotivation regards learners who were motivated but the motivation reduces later, and thus learners can be motivated again, which is described as remotivation (Thorner & Kikuchi, 2019). This study adopted the definition from Dörnyei and Ushioda (2021) that demotivation is “specific external forces that reduce or diminish the motivational basis of a behavioural intention or an ongoing action”.

2.2. Demotivating factors in English as a foreign language learning

Researchers have sought to identify demotivating factors of EFL learners. Sakai and Kikuchi (2009) focus on Japanese high school

students learning English with their 35-item demotivation questionnaire and identified five primary factors influencing demotivation: 1) teacher competence and teaching style, 2) instructional content and learning materials, 3) test scores, 4) lack of intrinsic motivations, and 5) inadequate school facilities. This foundational research laid the groundwork for subsequent research exploring the multifaceted nature of demotivation in language learning environments. Among these crucial demotivators, teacher behaviour was suggested as the least influential factor while the course difficulty best explained the demotivation (Kikuchi, 2015). Other researchers have been inspired and continue to expand more demotivators, such as learners' aptitude (learning difficulty), pressure from parents' high expectations for English learning (Song & Kim, 2017), influence from family members (Hassaskhah et al., 2015), and learners' negative attitude toward society's perceptions of English language (Krishnan & Pathan, 2013). Another contributing factor to demotivation is self-regulation, reflected in the learning process, from goal setting, behaviour monitoring, regulation and control, and cognition and motivation adjusting in the learning environment (Cho et al., 2020; Zhou et al., 2023).

In the study of demotivators in EFL learning, researchers typically categorise factors into external (contextual) and internal (learner-related) dimensions (Gao et al., 2022). This study also adopts these dimensions to classify previously defined factors (as shown in Table 1). However, there lacks an interplay between internal and external factors, since demotivation displays a negative process where learners' motivation declines due to both internal and external factors jointly (Gao & Liu, 2022).

The direct impact of demotivation on academic performance remains underexplored in motivation studies. While motivation has been wildly recognised as beneficial for academic success (Shang & Ma, 2024; Zeynali et al., 2019; Zhang & Wang, 2023), there is limited direct evidence supporting the hypothesis that demotivation negatively affects academic performance. One of the few studies indicated a negative correlation between demotivation and English test scores (Wang & Guan, 2020). Therefore, more empirical evidence is required to understand the direct impact of demotivation. Methodologically, there is a lack of mixed methods to develop a comprehensive and in-depth understanding of demotivation research (Gao & Liu, 2022).

2.3. Situated Expectancy-Value Theory (SEVT)

In motivation research, various theories have been employed to explore and interpret students' interest and motivation in learning activities. Most theories offer foundational insights that shape constructs related to positive emotions and psychological status in learning, such as self-efficacy (Bandura, 1977), growth mindset for success (Dweck, 2006) and achievement goals (Urduan & Kaplan, 2020). Expectancy-Value Theory (EVT) specifically highlights both positive and negative emotions associated with learning (Eccles, 1983; Wigfield, 1994; Wigfield & Eccles, 2020). EVT consists of three dimensions: 1) expectancy of success (ES), 2) subjective task values (STV)-which include intrinsic value, attainment value, and utility value, and 3) cost, which encompasses

Table 1
External and internal factors of demotivation

Dimensions	Factors	References
External (contextual) factors (Gao et al., 2022)	• Teacher competence and teaching style	• (Kikuchi, 2015; Sakai & Kikuchi, 2009)
	• Instructional content and learning materials	
	• Test scores	
	• Inadequate school facilities	
	• Pressure from parents' high requirements	
Internal (learners') factors (Gao et al., 2022)	• Lack of intrinsic motivations	• (Hassaskhah et al., 2015; Song & Kim, 2017) • (Kikuchi, 2015) • (Song & Kim, 2017) • (Krishnan & Pathan, 2013) • (Cho et al., 2020; Zhou et al., 2023)
	• Learners' aptitude (learning difficulty)	
	• Learners' negative attitude toward society	
	• Lack of self-regulation	

Table 2
Participants' demographic information (N = 426).

Variables		Categories	N	%
Gender		Female	338	79.3
		Male	88	20.7
Major		Clinical Medicine	108	25.4
		Stomatology	23	5.4
		Medical Imaging	32	7.5
		Nursing	252	59.2
		Bioengineering	3	0.7
		Pharmacy	8	1.9
English Proficiency	CET-4 < 425	Relatively low	216	50.7
	CET-4 ≥ 425	Intermediate	178	41.8
	CET-6 ≥ 425	Upper intermediate	32	7.5
Current English learning motivation		Very weak	108	25.4
		Somewhat weak	188	44.1
		Somewhat strong	111	26.1
		Very strong	19	4.5

N number of participants.

effort cost, emotional cost, and opportunity cost). The intrinsic value and utility value align with concepts of intrinsic motivation and extrinsic motivation (Eccles & Wigfield, 2020). Some researchers have suggested that cost should be treated as a singular dimension according to statistical analysis, proposing a renaming of the theory to EVC Theory- Expectancy, Value, and Cost (Barron & Hulleman, 2015). However, it was later argued that it was not all influencing factors need to be reflected in the theory's name (Wigfield & Eccles, 2020). As an advancement of this framework, EVT has been refined into SEVT (Situating Expectancy-Value Theory), emphasising the significant role of social context by adding a fourth dimension: 4) situated context (Eccles & Wigfield, 2024; Wigfield & Eccles, 2020). Consequently, the reformulated version indicates that culture and contextual factors significantly influence personal choices - a notion supported by empirical data (Muenks et al., 2023).

Situating Expectancy-Value Theory (SEVT) has significantly advanced our understanding of students' motivation and academic outcomes. However, existing empirical studies on motivation utilising SEVT primarily focus on STEM education contexts. Key areas of research include initial teacher education students in STEM (science, technology, engineering, mathematics) (Fielding et al., 2022), secondary school students (Adler et al., 2024), high school students (Jiang & Zhang, 2023; Ozturk et al., 2024; Zhang et al., 2023), college students leave in SETM major (Perez et al., 2014), and parental beliefs influence on youth math motivational beliefs (Ozturk et al., 2024). For example, Fielding et al. (2022) studied 41 Australian Initial Teacher Education (ITE) students engaging with three challenging mathematical tasks. Using pre- and post-surveys and focus group interviews, they found that most students had high expectations of success and believed they would understand the tasks. Their motivation was closely linked to their situational interest in the problem context, guided by SEVT. Similarly, Jiang and Zhang (2023) also examined the expectancy, value, and cost beliefs of Chinese high school students in math and English learning. Their findings revealed the multifaceted nature of student motivation across five dimensions and highlighted how cultural factors influence students' perceptions of learning costs. However, more research with SEVT framework focuses on middle school and high school students (Jiang & Zhang, 2023; Zhang et al., 2023) in STEM education, while few applications of SEVT have been explored in studying medical education, especially in EFL students' language learning.

Many SEVT studies tend to present successful stories by reporting a positive influence on students' motivation and outcomes (Eccles & Wigfield, 2024), often neglecting the importance of understanding demotivation. The rationale for understanding demotivation within the SEVT framework involves exploring the interplay between expectancy, value, cost, context and their relationship with demotivation. Researchers found that university dropout may be affected by the low level of expectancy (Schnettler et al., 2020), and decreased values of tasks

(Eccles & Wigfield, 2020; Wigfield & Eccles, 2020). Various costs, such as fear of failing in exams, can reduce the motivation in engaging learning activities (Kosovich et al., 2015). Situated context has not been studied thoroughly compared to other elements in SEVT. This study aims to understand undergraduate students' demotivation in English learning in medical education, SEVT is adopted as the theoretical framework for its comprehensive focus on both positive and negative factors and its growing relevance to the medical school context. Accordingly, the following research questions are proposed:

- 1) What factors contribute to Chinese medical students' demotivation in English learning?
- 2) How is the demotivation in medical students' English learning correlated with their academic performance?
- 3) How can the SEVT framework enhance understanding of demotivational factors in medical students' English learning experiences?

3. Methodology

3.1. Participants

A total of 426 second-year undergraduate students from a medical university in Southern China participated in the research as a convenience sample. These students were from six medical-related majors: Nursing, Clinical Medicine, Medical Imaging, Pharmacy, Stomatology, and Bioengineering. Of the participants, 338 (79.3 %) were female and 88 (20.7 %) were male. Their English proficiency was collected based on their performance on CET (College English Test).¹ The largest proportion of students (50.7 %) fell into the relatively lower-intermediate category (CET-4 < 425), while 178 (41.8 %) were at the intermediate level (CET-4 ≥ 425 but with CET-6 < 425), and 32 (7.5 %) were classified as upper-intermediate (CET-6 ≥ 425) (see Table 2). Data were collected at the end of their second academic year, and all participating students had completed their two-year “College English” course.

3.2. Instruments

The research collected both quantitative and qualitative data via a questionnaire with three sections. The first section included participants' demographical information, including gender, major, and current level of English learning motivation, rated from “1 = very weak” to “4 = very strong”.

The second section was adapted from the 35-item demotivation questionnaire developed by Sakai and Kikuchi (2009) and translated into the participants' first language (Chinese) with evaluation by an expert researcher. A pilot study was conducted with 47 students and two expert teachers/researchers. To better fit the current research context, certain dimensions and items were revised or removed. For example, the original dimension “Learning contents and materials” was divided into two subdimensions: *Teaching content* (e.g., “The courses mainly focus on translation/ grammar/ translation/ exam-oriented contents.”) and *Textbooks and material* (e.g., “The contents of the textbooks are not practical.”). Additionally, the item “The Internet was not used.” was removed since all the classrooms had access to the Internet at the time of data collection. In addition, the dimension of “Lack of time management and self-discipline ability” was added according to pilot feedback. In the final version, items with factor loading lower than 0.6 were removed, resulting in 22 items across seven dimensions retained in the second section of the current questionnaire with a 5-point Likert scale from 1

¹ The College English Test (CET), comprising Band 4 and Band 6, is a large-scale standardised examination sponsored by Chinese Ministry of Education and implemented by the Education Examinations Authority of the Ministry of Education. Only students who receive a score ≥ 425 in CET-4 are eligible to take the CET-6.

(strongly agree) to 5 (strongly disagree) (See Appendix A). The seven dimensions are teaching content, failure experiences of English learning, teacher-related factors, textbooks and teaching materials, school facilities, lacking goals of English learning, and lack of ability of time-management and self-discipline. The finalised questionnaire demonstrated good internal consistency (Cronbach's $\alpha = 0.844$) based on cleaned data.

The third section consisted of an open-ended question designed to explore participants' detailed feelings and behaviour related to their demotivation. As participation in the survey was voluntary, participants were encouraged (though not required) to provide any supplementary information on factors affecting their English learning motivation. There was no word limit for their responses, allowing for voluntary and comprehensive expression.

3.3. Data collection and analysis

The questionnaire was distributed through the online survey platform *Wenjuanxing* (<https://www.wjx.cn/>). 426 valid responses were collected after data cleaning, which excluded outliers such as responses submitted in less than three minutes and those with missing data. As for the ethical consideration, data collection commenced only after participants provided their consent by agreeing to the "Informed Consent" statement on the initial page of the questionnaire. This process ensured that the data collected were grounded in the informed consent of all participants. Participants were assured of anonymity of their responses, emphasising that the data would be used solely for research purposes and would be published accordingly. The entire research design adhered to ethical guidelines, and the study was conducted with the utmost respect for the participants' rights and confidentiality.

For data analysis, a mixed methods approach was adopted to explore the research questions. Quantitative data analysis involved descriptive analysis, Exploratory Factor Analysis (EFA), Confirmatory Factor Analysis (CFA), and correlation analysis have been conducted using SPSS 24.0 and AMOS 26.0. To address Research Question 1, EFA was employed to extract the major demotivating factors of English learning, and the dimensions were then confirmed fitness through CFA; to explore Research Question 2, correlation analysis was performed to investigate the relationship between students' English proficiency and the demotivators of their English learning; to further answer Research Question 3, SEVT framework was integrated into data presentation and discussion.

For qualitative analysis, qualitative data was collected from the open-ended question as a supplement to research question 1. Thematic analysis was chosen for its flexibility and capacity to further understand participants' perceptions of demotivating factors in their English learning (Braun & Clarke, 2006). A deductive (framework-driven) thematic analysis was employed, guided by the framework of factors extracted from the questionnaire validated through EFA and CFA, ensuring that the data were interpreted within the context of established constructs (Kiger & Varpio, 2020). While these predefined factors informed the coding, the thematic development allowed for flexibility, enabling the identification of additional categories that did not strictly conform to the original factor structure.

The data was analysed following the six-step procedure iteratively (rather than linearly) (Clarke & Braun, 2017; Kiger & Varpio, 2020). The coding template is presented in Table 3. To begin with, the authors read and re-read the data to perceive the overall meaning set off the data. Using the seven demotivating factors as a coding framework, any segments related to these factors or potentially linked to demotivation were highlighted and labelled as initial codes. The coding is idea-based, which means if certain words or phrases appeared repeatedly in one response, they were counted as "1" in frequency. If a single response reported more than one theme, each theme was counted separately. After the initial coding, codes that reflect similar or related content were categorised into one theme based on the framework, and the authors reviewed the data to avoid overlap between different themes. Emerging

Table 3

Coding scheme of qualitative data.

Initial coding labels	Themes	Codes	Example segments
Weak language foundation	Language foundation	LF	Weak foundation, hard, easy to forget, don't understand, difficulty, low scores, etc.
Low learning ability			Hard, easy to forget, don't understand, don't know how to improve, etc.
Heavy load of professional (medical) course learning	Heavy courseload	HL	Many courses, professional course pressure, lab reports, English move way to, etc.
Peer impact	Learning environment	LE	Learn alone, no speaking partner, etc.
Lacking communicating environment			No native English teachers, no real English communication context, etc.
- Classroom equipment - Teaching style/ pace	Teacher-related factors	TF	- Low temperature of air conditioner - Lack interaction, outdated teaching, fast/slow, exam-based/need exam strategies, etc.
- Personality and teaching beliefs - Previous teacher-related factor	Negative experiences	NE	- Ridicule mistakes, bored, lack encouragement - English teacher in junior school, previous English teacher
Previous exam-related factor			College entrance examination, previous scores, etc.
- Time management - Self-discipline	Low ability of self-regulation	SR	- No time, little time on English learning, etc. - Can't control myself from entertainment, no self-monitor, didn't insist, lazy, etc.
- Lack of interest in English learning - Reduced Importance	Lacking interest Others	LI O	- No interest/motivation, don't want to learn, etc. - Not as important as in high school

themes were elaborated accordingly. Disagreements were discussed with a more experienced researcher to reduce discrepancy and enhance dependability and consistency (Campbell et al., 2013). The quality of the qualitative analysis and credibility were established through the trust-building between the first author and participants, the informed consent on the first page of the questionnaire, and the reflexivity and refinement after the pilot analysis. The result of coding is presented in Section 4.4 with Table 10.

4. Results

4.1. Descriptive analysis of factors contributing most to the demotivation

The descriptive analysis of the tailored demotivation questionnaire (see Table 4) revealed negative skewness coefficients. This leftward shift indicates that the majority of participating medical students tend to agree with the demotivating factors presented in the questionnaire with most ratings clustering around the lower end of the scale (closer to 1). This trend suggests a significant presence of these demotivating factors among the participants.

Mean values for each item identify the extent of demotivation, with lower values indicating stronger demotivation (since 1 stands for "strongly agree"). The top five items that contribute to English learning demotivation are shown in Table 5. The items belong to three factors, including weak ability in time management (Q35, $M = 2.44$, $SD = 1.04$), the lack of self-discipline (Q34, $M = 2.58$, $SD = 1.16$), weak foundation in English language (Q8, $M = 2.67$, $SD = 1.13$; and Q6, $M = 2.74$, $SD =$

Table 4
Descriptive statistics of structural equation model variables ($N = 426$).

Items	Min	Max	M	SD	Skewness		Kurtosis	
					Statistic	Std. Error	Statistic	Std. Error
Q2	1	5	2.8920	1.037	-0.139	0.118	-0.310	0.236
Q3	1	5	3.1362	1.069	-0.320	0.118	-0.303	0.236
Q6	1	5	2.7441	1.109	-0.021	0.118	-0.874	0.236
Q7	1	5	3.0986	1.210	-0.230	0.118	-0.911	0.236
Q8	1	5	2.6714	1.129	0.130	0.118	-0.855	0.236
Q9	1	5	3.2700	1.214	-0.371	0.118	-0.817	0.236
Q10	1	5	3.0164	1.361	-0.171	0.118	-1.238	0.236
Q12	1	5	3.9906	0.985	-0.813	0.118	0.249	0.236
Q14	1	5	3.4953	0.946	-0.146	0.118	-0.262	0.236
Q16	1	5	3.5751	0.941	-0.474	0.118	0.124	0.236
Q17	1	5	3.2934	1.029	-0.286	0.118	-0.296	0.236
Q18	1	5	3.0329	1.118	-0.197	0.118	-0.672	0.236
Q20	1	5	2.8826	1.076	-0.095	0.118	-0.648	0.236
Q22	1	5	2.9437	1.018	-0.196	0.118	-0.477	0.236
Q25	1	5	3.9507	1.015	-0.809	0.118	0.138	0.236
Q26	1	5	3.4671	1.280	-0.431	0.118	-0.901	0.236
Q27	1	5	3.7770	1.038	-0.672	0.118	-0.065	0.236
Q29	1	5	3.7535	1.159	-0.747	0.118	-0.201	0.236
Q30	1	5	3.6737	1.099	-0.629	0.118	-0.269	0.236
Q31	1	5	4.1479	0.965	-1.105	0.118	0.980	0.236
Q34	1	5	2.5775	1.160	0.401	0.118	-0.600	0.236
Q35	1	5	2.4437	1.042	0.552	0.118	-0.136	0.236

1.11), and impractical contents in textbooks (Q20, $M = 2.76$, $SD = 1.07$).

4.2. Results of EFA and CFA on dimensions of English learning demotivation

To address Research Question 1, which examines the demotivating factors influencing medical students' English learning, both Exploratory Factor Analysis (EFA) and Confirmatory Factor Analysis (CFA) were conducted. This approach was necessary as the questionnaire had been adapted and required further validation to confirm its reliability and validity.

4.2.1. Results of the exploratory factor analysis

Two rounds of exploratory factor analysis (EFA) were conducted to investigate the dimensions of English learning demotivation. The initial EFA yielded a Kaiser-Meyer-Olkin (KMO) measure of sampling adequacy of 0.882 and a highly significant Bartlett's test of sphericity ($p < 0.001$), extracting ten factors. However, two factors were excluded due to underrepresentation, as they were associated with only one item each (Q1 and Q21), one factor ("English learning environment") and items with factor loading lower than 0.6 were removed in the final scale. A subsequent factor analysis confirmed seven significant factors, with a KMO value of 0.821 and a significant Bartlett's test ($p < 0.001$).

Table 6 presents the factor loading matrix data obtained from principal component analysis with varimax rotation. Seven effective factors were extracted. Each factor was summarized and named based on the content of the corresponding items with a reliability analysis (Cronbach's α values were presented). The items within the six factors demonstrated high reliability, but not within Factor 1 ($\alpha = 0.690$), which exhibited a relatively low internal consistency ($\alpha < 0.700$).

Table 5
Five items with most contribution to demotivation of English learning.

Item	Description	M	SD
Q35	Lack of preview and review, making subsequent English learning difficult.	2.44	1.04
Q34	Numerous distractions (e.g., phones and computers) beyond self-control.	2.58	1.16
Q8	Lack of knowledge on how to self-study English.	2.67	1.13
Q6	Difficulty in memorising English words and phrases.	2.74	1.11
Q20	The contents of the textbooks are not practical.	2.76	1.07

(Tavakol & Dennick, 2011).

Valid demotivating factors from EFA were Factor 7 ($M = 2.51$), Factor 4 ($M = 2.95$), Factor 2 ($M = 2.96$), Factor 1 ($M = 3.01$), Factor 3 ($M = 3.49$), Factor 5 ($M = 3.73$), and Factor 6 ($M = 3.86$). Factors 7, 4, 2 and 1 showed stronger demotivation over the total mean value of the questionnaire ($M = 3.23$), indicating a significant contribution to medical students' English learning demotivation. Factor 7 refers to the lack of ability of time management and autonomous learning, mainly the lack of preview and review, resulting in difficulty in subsequent learning. Factor 4 comprises teaching materials, characterized by uninteresting texts and outdated or less practical cases in English textbooks. Factor 2 refers to negative English learning experiences, including weak foundations from neglecting English in middle and high school, resulting in difficulty understanding lectures, low exam scores in past English learning experiences, lack of confidence, and difficulty in memorising vocabulary. Factor 1 relates to decreased motivation caused by translation- and grammar-based content in class. The latter three factors are mainly external influencing factors to demotivation, while Factor 7 relates to individual ability and self-discipline.

4.2.2. Results of the confirmatory factor analysis

To confirm the validity of the construct of the extracted seven factors from medical students' demotivation in English learning, CFA was conducted with AMOS 26.0 (Fig. 1). Table 7 offers the goodness of fit model which meets the specified criteria. The ratio of CMIN-DF is 2.048, GFI = 0.926, CFI = 0.942, PNFI = 0.727, TLI = 0.928, and RMSEA = 0.050, indicating a satisfactory model fit indices ($\chi^2/df \leq 3$, GFI ≥ 0.90 , CFI ≥ 0.90 , TLI ≥ 0.90 , PNFI > 0.50 , RMSEA ≤ 0.08) (Kline, 2016).

The convergent validity and discriminant validity of demotivating factors are reported in Table 8. It can be seen that correlations between factors were weak or mediate ($r < 0.7$), only correlations between failure experiences/ weak foundation and lacking goals of English learning had a relatively strong correlation ($r = 0.609$). All correlation coefficients were lower than the square root of the average variance explained (AVE) for each factor, indicating the discriminant validity. The composite reliability (CR) for six factors was above the value of 0.7, and slightly below 0.7 for Factor 7 (0.698), acknowledging the convergent validity of the model.

In addition to the self-reported factors, a correlation analysis was conducted to explore whether medical students' demotivation in English learning is related to their English proficiency at different stages.

Table 6

Rotated component matrix of seven factors (22 Items).

Items	Rotated component matrix ^a						
	Component						
	F1	F2	F3	F4	F5	F6	F7
Factor 1: Teaching content ($\alpha = 0.70$, $M = 3.01$)							
Q3	0.857	0.149	0.085	0.098	0.046	0.098	−0.036
Q2	0.815	0.021	0.167	0.178	0.056	0.126	0.127
Factor 2: Failure of English learning or weak foundation ($\alpha = 0.69$, $M = 2.96$)							
Q7	0.061	0.838	−0.025	0.021	0.049	0.100	0.057
Q10	0.054	0.820	0.014	−0.052	0.011	0.145	0.061
Q9	0.032	0.739	0.017	0.151	0.110	0.243	0.085
Q8	−0.005	0.697	0.052	0.048	0.041	0.126	0.261
Q6	0.061	0.689	0.025	−0.021	0.032	0.115	0.062
Factor 3: Teacher-related factors ($\alpha = 0.84$, $M = 3.49$)							
Q17	0.028	0.058	0.815	0.221	0.059	0.026	−0.002
Q14	0.099	−0.031	0.797	0.145	0.066	0.059	0.127
Q16	0.056	0.056	0.767	0.111	0.151	−0.027	0.123
Q12	0.147	−0.018	0.602	0.032	0.249	0.221	−0.254
Factor 4: Textbooks and teaching materials ($\alpha = 0.78$, $M = 2.95$)							
Q22	0.090	−0.018	0.098	0.861	0.071	0.083	0.079
Q20	0.094	0.068	0.121	0.844	−0.009	0.100	0.034
Q18	0.122	0.050	0.352	0.741	−0.003	0.023	0.087
Factor 5: Teaching facilities or equipment ($\alpha = 0.76$, $M = 3.73$)							
Q27	0.067	0.112	0.091	0.171	0.824	0.049	−0.001
Q26	−0.001	0.018	0.106	−0.036	0.823	0.034	0.014
Q25	0.039	0.063	0.169	−0.057	0.769	0.029	0.068
Factor 6: Lacking goals of English learning ($\alpha = 0.82$, $M = 3.86$)							
Q30	0.056	0.246	0.054	0.032	0.062	0.829	0.069
Q31	0.093	0.143	0.075	0.088	0.044	0.796	0.065
Q29	0.122	0.401	0.055	0.116	0.014	0.733	0.111
Factor 7: Lack of ability of time management & autonomous learning ($\alpha = 0.70$, $M = 2.51$)							
Q34	0.083	0.135	−0.009	0.168	0.078	0.208	0.822
Q35	0.012	0.349	0.127	0.024	0.013	0.016	0.784

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

^a Rotation converged in 6 iterations.

4.3. Correlation analysis

To address Research Question 2, this study conducted a correlation analysis between participants' English proficiency (CET scores) and the results of the demotivation questionnaire. Utilising Spearman correlation analysis, the study identified varying degrees of correlation between the demotivating factors identified in the questionnaire and the students' scores, as shown in Table 9.

The correlation analysis revealed that medical students' English proficiency was moderately ($0.4 \leq r \leq 0.7$, $p < 0.05$) or weakly ($0 < r < 0.4$, $p < 0.05$) correlated (Field, 2024) with their demotivation of English study. Among the 12 items correlated with English proficiency, two items in bold (Q7 and Q10), which belong to Factor 2 (Failure experience and weak foundation), were reported moderately with their CET scores. This suggests that students with low English proficiency were more likely to experience high levels of demotivation due to negative or failed English learning experiences. However, no items demonstrated a strong correlation, suggesting that English proficiency alone is not a decisive factor in demotivation among medical students.

4.4. Qualitative data analysis of demotivating factors

Qualitative data collected from the open-ended voluntary question in the questionnaire consisted of 91 valid responses out of 114 after data

cleaning, focusing on English learning demotivation. As there was no specific word count requirement, most responses were brief, typically consisting of one to two sentences or a few keywords, resulting in a total word count of 2609 Chinese characters. Despite the brevity of the responses, the concise nature of the Chinese language did not impede students from effectively conveying their intended meanings. Responses that were irrelevant to demotivation, such as comments like “Nothing” or other unrelated comments, were excluded from the analysis. As presented in Table 10, these responses offer deeper insights and more detailed explanations of the demotivating factors identified in the study. Since three responses reported more than one theme, the number of mentioned themes reached 94.

The qualitative analysis contextualises and elaborates on the quantitative findings, providing a comprehensive understanding of the underlying causes of demotivation in medical students' English learning from the following dimensions.

4.4.1. Weak language foundation and low academic ability

The factor that emerged most frequently mentioned by participants was “language foundation”, which included a low level of aptitude and difficulty in English learning. Many medical students perceived English as “too difficult to learn”, reporting difficulties in memorising words and grammar rules since they “always forget the vocabulary meaning although had memorised every day”. Specific learning strategies were

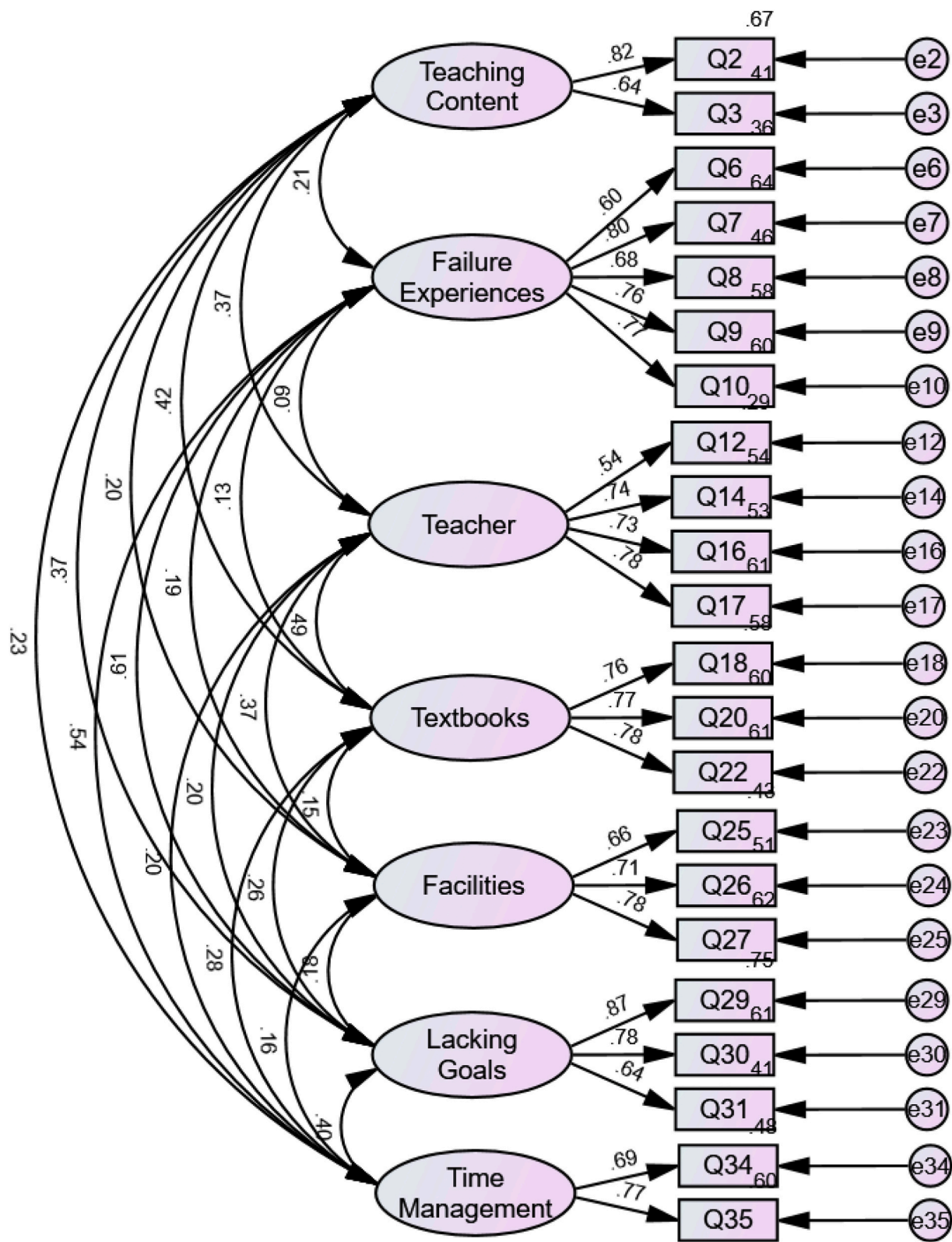


Fig. 1. The final CFA model of medical students' demotivation of English learning.

also problematic: students reported, “[I] just don't know how to improve listening”. These challenges were attributed to their “poor foundation”, indicating that inadequate prior knowledge of English significantly contributes to their demotivation. This reflects a low expectancy of success in English learning, as students believe that their weak foundational English language knowledge hampers their future learning

opportunities.

4.4.2. Heavy workload in medical courses

The second most frequently cited demotivating factor was the significant burden imposed by the heavy workload of medical courses. Students emphasised that their rigorous schedules and demanding

Table 7
The goodness of fit estimation.

Criteria	Threshold			Evaluation
	Terrible	Acceptable	Excellent	
CMIN	384.942			Excellent
DF	188.000			
CMIN/DF	2.048	>5	>3	
RMSEA	0.050		< 0.08	Acceptable
GFI	0.926	>0.80	>0.90	
CFI	0.942	>0.80	>0.90	
PNFI	0.727		>0.50	Acceptable
TLI	0.928	>0.80	>0.90	Acceptable

coursework left them with insufficient time and energy for English learning. The pressures to master subject knowledge through preparation, review, and assignments, lead them to “prioritise professional courses over English studies”. In addition to theoretical courses, they also have experimental courses, which require them to complete lab reports to carefully record the objectives, methods, processes, results and other relevant aspects according to each experiment. One student expressed “We have very thick textbooks to understand and memorise, and many lab reports to submit. I have little time for English.” This was echoed by others who mentioned the overwhelming number of assignments, projects, and quizzes in their medical studies. This can be considered as both the effort cost and the opportunity cost since students realised that dedicating time to English learning could detract from their medical assignments.

4.4.3. Learning environment and peer impact

Responses related to peer influence, a lack of English communication environment, and inadequate classroom facilities comprised this category. Limited interaction with native English speakers and peer dynamics negatively affected the overall learning community. Poor facilities also made students uncomfortable during English classes. The perception indicates a relatively low expectancy of English learning, as they believe that low-quality school facilities and a passive learning atmosphere among peers diminished their motivation. This reflects the significance of the classroom contextual factor in shaping students' expectations and demotivation toward learning English.

4.4.4. Teacher-related factors

Teacher-related factors focused on teaching methods and teachers' attitudes of specific teachers toward students. Students reported that teachers' inadequate encouragement from teachers or ridiculed for making mistakes demotivated them. This presents a psychological cost or emotional cost for students since they may experience uncomfortable in English classes due to negative teacher attitudes. Interestingly, conflicting perceptions emerged regarding teaching pace; some students felt that “the teacher taught too fast and hard to catch up”, while another reported that “the teacher taught slowly” and that made him “boring in class”.

Table 8
Convergent validity and discriminant validity of each factor.

Factors	Convergent validity		Discriminant validity						
	AVE	CR	1	2	3	4	5	6	7
1. Teaching content	0.542	0.700	0.736						
2. Failure experience and weak foundation	0.528	0.847	0.212	0.726					
3. Teacher-related Factors	0.491	0.791	0.372	0.095	0.700				
4. Textbooks	0.595	0.815	0.421	0.127	0.492	0.771			
5. Teaching facilities and equipment	0.519	0.763	0.196	0.192	0.368	0.153	0.720		
6. Lacking goals of English learning	0.590	0.809	0.365	0.609	0.200	0.259	0.177	0.768	
7. Time management	0.537	0.698	0.233	0.540	0.195	0.278	0.155	0.399	0.733

CR: Composite Reliability, AVE: Average Variance Extracted.

Table 9
Correlation analysis of English proficiency and demotivators of English learning (N = 426).

Factors	Items	r
Factor 1: Teaching content	Q2	0.069
	Q3	0.084
	Q6	0.296**
Factor 2: Failure experience and weak foundation	Q7	0.451**
	Q8	0.392**
	Q9	0.375**
	Q10	0.489**
	Q12	−0.052
Factor 3: Teacher-related Factors	Q14	−0.023
	Q16	0.011
	Q17	−0.058
	Q18	−0.040
Factor 4: Textbooks	Q20	−0.106*
	Q22	−0.137**
Factor 5: Teaching facilities and equipment	Q25	0.081
	Q26	0.082
	Q27	0.072
Factor 6: Lacking goals of English learning	Q29	0.324**
	Q30	0.279**
	Q31	0.206**
	Q34	0.262**
Factor 7: Time management	Q35	0.329**

** . Correlation is significant at the 0.01 level (2-tailed).

* . Correlation is significant at the 0.05 level (2-tailed).

4.4.5. Negative past experiences in English learning

This category referred to students' negative experiences from past English learning contexts, including low academic performance in previous exams (such as middle school exams/college English entrance examinations), and aversion to English due to verbal discouragement from elementary and middle school English teachers. This is another teacher-related factor but more of an impact from previous negative feelings on current motivation. These experiences contribute to low expectancy and psychological costs with upset feelings.

4.4.6. Lack of self-regulation

The “lazy” appeared seven times highlighting the lack of self-regulation ability. Some participants attributed their unsatisfactory with “time management” to either personal ability or the burden of the medical course schedule. Thus, this factor is closely linked to *heavy workload of medical courses*. Students reflected that they “have many medical courses and lab reports to complete”, and “have little time and energy to allocate to English study” due to “inertia”. The psychological cost manifests as anxiety and struggle stemming from an imbalance in prioritising medical studies over English learning. Accordingly, this also leads to low expectancy of English learning since they realise that learning and practising English requires a large amount of energy and time.

Table 10

Coding results of qualitative data for demotivating factors.

Themes	Codes	Example quotes	N	%
1. Language foundation	LF	"My English is always poor, it's really hard for me to understand English courses."	27	30 %
2. Heavy courseload	HL	"We have very thick textbooks to understand and memorise and many lab reports to submit, I have little time for English."	17	18.7 %
3. Learning environment	LE	"The English learning atmosphere is weak, not many students make an effort to learn English." "The projector was old."	14	15.4 %
4. Teacher-related factors	TF	"The teacher sometimes emphasises students' pronunciation, I don't like this."	10	11.0 %
5. Low ability of self-regulation	SR	"I am weak in self-control"; "I always forget after-class exercises and spend much time on campus activities and phone games."	9	9.9 %
6. Negative experiences	NE	"My English has been poor since primary school."; "I have no confidence in English learning."	8	8.8 %
7. Lacking interest	LI	"I just have no interest in learning English and I don't think I will need it in the future."	6	6.6 %
8. Others	O	"College English courses are not as important as it was in high schools."	3	3.3 %

N number of participants mentioning the codes.

4.4.7. Lack of interest in English learning

Some students expressed a lack of relevance or necessity for learning English for their future endeavours or have no interest in English learning. Some directly expressed "I just have no interest in learning English." or "I don't think English learning is useful.". This shows a low level of intrinsic value of English learning, as well as a lack of goal-setting in English learning regarding its importance.

4.4.8. Others - reduced importance and difficulty of college English courses

Responses categorised under "Others" indicated perceptions beyond the main themes identified earlier. For instance, some students reported that "English courses in college are not as important as they were in high school" since university courses had less intensity and fewer stringent requirements. Additionally, some students found the final exams "too simple", leading to disinterest. This perceived lack of challenge may further demotivate students. This factor reveals the low utility value of learning English that is overlooked regarding its relevance now and in future careers.

The heavy workload in medical courses emerged as a noted demotivator which was not covered in the initial demotivation questionnaire. Most qualitative themes aligned with the questionnaire findings, while this significant demotivator was emphasised by medical students. Considering that the voluntary qualitative responses reflect authentic subjective feelings, further investigation into course arrangement confirmed this finding. The medical curriculum requires substantial time investment for lectures, lab sessions and report completion. For instance, professional medical courses during the second year accounted for an average of 20.4 class hours per week compared to only 4.0 class hours per week for College English (see Appendix B). This stark contrast highlights the disproportionate time allocation and the prioritization of medical courses over English learning. Consequently, this can partially explain students' perception of a "heavy load of medical courses" and time-management challenges.

5. Discussion

In exploring the demotivation experienced by medical students in EFL learning at a Chinese medical university, this study sheds light on diverse facets that contribute to students' perspectives.

5.1. Main demotivators in English learning for medical students

To begin with, descriptive analysis reveals that the main demotivators in medical students' English learning involve both internal factors, such as the lack of self-regulation ability, and external factors, such as the heavy load of professional coursework.

Besides, factor analysis uncovers three valid and significant demotivators out of seven. Medical students reported both internal factors (weak ability in time management and weak foundation in English language textbooks) and external factors (impractical contents in

textbooks) as key contributors to their demotivation in English learning. Teacher-related factors, teaching facilities or equipment, and the lack of English learning goals were valid factors, but not as important as the aforementioned three. Teachers' teaching methods were reported but not the primary demotivators, which aligns with previous research findings (Sakai & Kikuchi, 2009). This suggests that most of the teachers in the current case regulate their emotions and behaviour well, engaging students in classroom learning efficiently and fostering the motivation for English learning (Liu & Chu, 2022). This finding lays the groundwork for potential interventions and remotivating strategies to meet their needs (Thorner & Kikuchi, 2019).

A novel finding was the identification of the heavy workload associated with medical courses as a significant demotivator. The demanding nature of medical education leaves students with limited time and energy to engage in additional academic pursuits, such as language learning. Medical students are particularly susceptible to stress, burnout and even mental disorders than other groups, which can exacerbate feelings of demotivation (Jumart et al., 2020; Shrestha et al., 2021; Stirparo et al., 2024). This unique demotivator faced by medical students is often overlooked in language learning motivation research. While previous research has focused on EFL demotivation among technological university students (Li, 2021), this study provides broader insights into situated medical educational contexts.

Another noteworthy demotivator of medical students' English learning is the lack of self-regulation, which includes unclear goal setting, insufficient self-monitoring of autonomous learning, and inadequate cognitive adjustment on EFL learning. This, combined with the heavy load of medical courses, is perceived by students as "laziness" in qualitative data. Further, students attributed their absence of goal-setting and targeted plans for English learning to their inadequate time management competencies. This finding sharply contrasts with the positive outcomes associated with strong goal-setting and self-regulation abilities, which are linked to better academic performance and higher motivation levels (Liu, 2022). Moreover, the lower level of self-regulation ability among demotivated medical students in the university environment may relate to the constant pressure (Thorner & Kikuchi, 2019), such as the high pressure of high-stakes exams like the university entrance examination in China and the intensive schedule in medical education, as indicated by this research.

Accordingly, teachers can strengthen students' time-management skills and emphasise the importance of long-term goal-setting. In addition to building vertical professional knowledge and practical abilities, proficiency in English is crucial for the development of horizontal abilities. These include perceiving the latest research trends from international literature, publishing findings in international journals, engaging in international conferences and communicating with experts and peers in the fields. Therefore, medical students should strive for the concurrent advancement of professional and English abilities for lifelong learning.

5.2. Correlation between demotivators and English proficiency

Although a relationship has been reported between medical students' demotivation and their English proficiency, no strong correlation has been identified. This challenges existing research that posited language proficiency as a significant contributor to learners' demotivation (Danesh & Shahnazari, 2020; Gao et al., 2022; Sakai & Kikuchi, 2009; Zeynali et al., 2019). The present findings suggest that English proficiency alone does not necessarily determine the extent of demotivation, although it moderately correlates with one demotivation factor, namely, struggles in English learning stemming from an inadequate foundational understanding.

The data show that English proficiency, when considered independently, does not conclusively predict the degree of demotivation. The observed moderate-to-weak relationship highlights the importance of other specific demotivators over mere proficiency scores. This broader perspective suggests that addressing demotivation requires a comprehensive approach that considers various internal and external influences, such as certain personality traits (Zhang & Wang, 2023), rather than concentrating exclusively on proficiency levels. Understanding this multifaceted relationship allows for more effective interventions designed to mitigate demotivation. Accordingly, although language proficiency is significant, more interventions need to be taken to reduce demotivation by diminishing the sense of failure and enhancing the expectancy of success in English learning.

5.3. Understanding demotivation from SEVT perspectives

Insights into medical students' demotivation are discussed under three dimensions: 1) expectancy of success (ES), 2) subjective task value (SVT) (including attainment value, intrinsic value, and utility value), and 3) cost and situated context (medical education).

5.3.1. Expectancy of success (ES)

Students' expectancy of success in English learning reflects their beliefs about potential performance in English learning tasks. Our findings indicate that factors such as a "weak foundation", "negative English learning experiences", "low ability for self-regulation" and an unsatisfied "learning environment" (classroom contextual factors) contribute to a low level of expectancy, further reducing their motivation. These factors are mainly internal, and highlight participants' lack of confidence in their ability to succeed in English learning as they face difficulties with basic grammar and vocabulary. Students who have had previous unsatisfactory experiences of English learning may believe they are not capable of learning English. Although previous studies have suggested that high expectancy can enhance students' engagement in tasks (Fielding et al., 2022) and improve their motivational state (Ghorbani & Bund, 2020), the present research provides further evidence from a negative perspective. This critical insight emphasises the importance of addressing these internal barriers to boost students' confidence and expectancy of success in English learning.

In terms of the implication in this point, even relatively minor successes can have a positive impact on enhancing their English self-efficacy (ES), which is lacking for medical students with lower English proficiency. Examples of such minor achievements might include completing a challenging English assignment, creating English-language Vlogs to document and share their campus life or local culture, responding to questions in English in front of classmates, or achieving a slightly better score on an English test from the previous attempt, etc. These seemingly trivial successes can cumulatively build and reinforce an individual's belief in their English language capabilities.

5.3.2. Subjective task value (STV)

In SEVT, STV reflects individual value in engaging in tasks (Eccles, 2005). Attainment value, intrinsic value, utility value, and cost can be observed in medical students' responses to demotivating factors. An

unexpected finding is that although participants have a clear understanding of their identities as medical students, they assign minimal value to English language learning. Some students therefore reported a lack of interest in English, indicating low intrinsic value. Despite recognising their future professional roles as doctors, nurses, dentists, pharmacists, or researchers, they show a limited understanding of the importance and usefulness of English learning in both their medical school life and future careers. This supports the argument that decreased values may reduce motivation for learning engagement (Eccles & Wigfield, 2020; Wigfield & Eccles, 2020).

5.3.3. Cost and situated context of medical education

The findings reveal a deep interplay between demotivators and other SEVT dimensions, specifically cost and the situated context of medical education. Regarding cost, student demotivation is influenced by the psychological cost, effort cost, and opportunity costs associated with English language learning. Factors such as "teacher-related factors" (such as discouraging teaching feedback, teacher-centred teaching style, and teachers' negative attitudes), "negative past learning experiences", "lack of self-regulation" and "heavy workload" highlight these costs. Students feel discouraged by the lack of teacher encouragement and low scores, contributing to a negative perception of English learning. This provides evidence to previous research that teacher burnout can impair the energy to inspire and motivate their students (Liu & Chu, 2022). However, high stress and turnover intentions hinder teachers' ability to maintain a supportive learning environment, leading to greater student demotivation and negative views of English learning (Liu, 2024). Both students' and teachers' wellbeing require more attention from key stakeholders. Additionally, students with poor self-regulation often attribute their struggles to "laziness," reflecting difficulties in time management and energy allocation. They recognise the importance of English for their grade point average (GPA) and future global communication but blame themselves for not dedicating enough time. The time and effort required for English learning detract from their medical studies, challenging existing research that views this cost as an opportunity for improvement (Fielding et al., 2022).

Furthermore, this study explores English learning demotivation within the situated context of medical education. To our knowledge, this dilemma has not been reported in previous research. Existing empirical studies explore the relationship between students' motivation and their expectancy, values and cost in Math and English learning tasks with positive results (Adler et al., 2024; Jiang & Zhang, 2023; Ozturk et al., 2024). The current research extends the application of SEVT to undergraduate medical students, offering new insights into English learning demotivation within the medical educational context. A notable demotivator identified in the present research is the heavy workload associated with medical courses. Qualitative data analysis revealed that students perceive the "heavy workload" as a barrier to allocating sufficient time for English learning, viewing English and their professional coursework as conflicting tasks. Students acknowledged the importance of both tasks but prioritized medical course learning over English by attributing their demotivation to a lack of self-discipline and "laziness".

5.4. Implication in remotivating medical students' English learning

To remotivate demotivated medical students in their EFL learning, it is essential to understand their needs and perceptions of demotivation, as it may develop from deficiencies in learners' needs (Thorner & Kikuchi, 2019). Interventions of remotivation can be addressed from different dimensions of SEVT (enhancing expectancy and value and decreasing cost). First, to enhance the expectancy of success in English learning, teachers' intervention can begin by facilitating students' goal setting (Wang & Guan, 2020; Zeynali et al., 2019) and helping them create targeted learning schedules for better time management. Meanwhile, frequent positive feedback on minor success at each step can enhance their expectancy of success (Liu et al., 2021) and reduce the

mental cost of achieving huge and ultimate success in English learning, which is lacking for medical students, particularly with the ones with lower English proficiency. This reflects a deep interplay between internal and external demotivating factors. This preparation is also essential for future professional medical academic communication, such as literature publishing and attending international conferences (Cheng, 2016).

Second, to foster medical students' intrinsic and utility value in English learning, more medical-related learning materials and tasks can be assigned to enhance their willingness to engage and communicate. Academic achievement can serve as a way to remotivation (Leksuwankun et al., 2023). In this case, lecturers and tutors can adapt textbooks or redesign tailored materials to engage medical students in English learning (Guan & Scott, 2024). Extracurricular activities with authentic communicative involvement in English and learning technology integration are encouraged (Guan & Scott, 2024; Wang et al., 2024), such as collaborative academic reading on the latest medical findings, mock academic conference presentations or paper writing, or creating vlogs about medical school life and experimental processes. Well-designed classroom activities, which are closely related to medical students' future careers, are expected to raise their awareness of the value of English learning and thereby enhance their motivation.

Lastly, to reduce the emotional cost of English learning in medical education, actions can be taken to release anxiety and the lack of confidence in English learning. For learners with low achievement, whether with a "weak foundation" or "negative experiences", anxiety and confidence deficit contribute to their English learning demotivation (Falout et al., 2009), which are proven mediated by learning buoyancy (Liu et al., 2024). Concerning confidence empowerment and resilience cultivation, researchers emphasised that teachers play a significant role (Liu et al., 2024; Liu & Chu, 2022). By modelling resilience and providing support, teachers can help medical students develop language knowledge and skills necessary to overcome challenges, which can enhance their motivation to learn English. L2 learners' belief that language ability is fixed (cannot be improved) may lead to demotivation (Albalawi & Al-Hoorie, 2021). Therefore, it is vital to not only build confidence in medical students in minor success as previously mentioned but also cultivate a growth mindset that failure at the moment can be treated as a chance to improve, rather than a summative sentence to end their English learning journey. Interventions that can enhance buoyancy are thus expected to mitigate students' emotional challenges, such as anxiety (Liu et al., 2024). Although English learning motivation may not undergo a drastic transformation, interventions (teachers' behaviours, for example) can have a slow, positive effect over time (Dewaele et al., 2022).

6. Conclusion

Guided by the Situated Expectancy-Value Theory (SEVT), this study reconceptualises motivation in EFL learning among Chinese medical students by uncovering the complex interplay of internal and external demotivators. Internal factors, such as self-regulation challenges, and external factors, including the demanding workload of medical education, were identified as key demotivators. This research contributes by reconceptualising motivation through SEVT dimensions in English language learning. The study reveals expectancy of success, subjective task value, and perceived cost underscore the significance of educational interventions. These interventions aim to bolster students' self-assurance and recognise the value of English in their future medical vocations, while also addressing the emotional and effort costs associated with language learning. The findings suggest the importance of context in understanding demotivation and call for future research to explore longitudinal patterns and the development of effective strategies for remotivating students within the unique pressures of medical education.

7. Limitation and implication

This study has limitations that should be addressed in future research. First, focusing on a single medical university in Southern China may limit the generalizability of the findings. Future research could explore demotivation in diverse medical education contexts to provide a more comprehensive understanding. Second, while the cross-sectional design offers insights into medical students' demotivation in EFL learning, a longitudinal approach could provide a more comprehensive and dynamic view of demotivation, recognising that it is not static but a fluid cognitive state. Finally, future studies could test the effectiveness of the remotivation strategies proposed in this study to regain motivation among medical students. Investigating effective interventions and support mechanisms can facilitate the development of programs aimed at enhancing student motivation and improving language learning outcomes.

CRedit authorship contribution statement

Xiaoting Ji: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **April Jiawei Zhang:** Writing – review & editing, Writing – original draft, Methodology, Conceptualization.

Ethics statement

Ethical review and approval were not required for this study, as it did not involve any medical procedures, human clinical trials, or research on laboratory animals. The research utilised non-interventional methods in accordance with local legislation and institutional requirements (*Statement on non-requirement of ethical review* has been obtained from Institutional Review Boards of the first author's affiliation). Informed consent was obtained from all participants included in the study.

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Declaration of competing interest

The authors report there are no competing interests to declare.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.actpsy.2025.104716>.

Data availability

Data will be made available on request.

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